



# Proposed CEA Link — December 2026

Cover at Seventeen with Saturday rebuilt — nine turns down to four and six start times down to three, weighted toward the Saturday late for Wembley

Prepared 22 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules · figures on this page are computed, not typed

## PROPOSAL

### Saturday Four

Saturday rebuilt - four turns, three start times, weighted to the late for Wembley

CODE **S4-24-EXT**

FINGERPRINT **481ba9ed**

BUILT FROM duty table — · seed — · 24 lines · 4 cover weeks

LINEAGE The FIFTH revision in one line, and the first to change a DUTY rather than the order of the weeks: "Weeks 17-18 Swapped" (WS-24-EXT · 0bebb675) → "Targeted Fatigue Redo" (TF-24-EXT · 8eef9a13) → "Three Mondays" (TM-24-EXT · fe90c0b8) → "Cover at Seventeen" (C17-24-EXT · edc1b731) → this, whose Saturday table was searched from scratch on Saturday's own minute budget. Monday to Friday and Sunday are carried over cell for cell.

✓ **24** lines designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

⚠ **3** fatigue factors present (today: 5)

## 10 of 18

shift times are today's

8 times are new; the rest are turns people already work

## 9 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

## 2 of 20

working weeks are one turn

same clock time all week, as 7 of today's 16 are

## 5 in 24

full weekends off

21% of the rotation, against 20% today (4 in 20)

## 4 cover weeks

lines 1, 7, 12, 17

gaps of 6, 5, 5, 8 around the wheel — unchanged; this revision touched Saturday only

## 3h 54m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

## What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

It was not drawn by this app. It arrived as a Word table — a proposal from outside the workspace — and was read straight into the app's own shape, then judged by the same modules that judge a design the workspace generated itself: runDesignChecks, weeklyHours, assessHardLimits, assessFatigue, scoreOrder and calcHourlyCoverage. There is no search behind it and nothing to reproduce; what can be checked is every figure on these pages, each computed from the cells on page 6. Paste those cells into Links → Import and the workspace restates all of it.

**What it is not.** Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 5 and 6 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

## THE ROTATION

## The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

| LINE  | SUN            | MON            | TUE            | WED            | THU            | FRI            | SAT            | MON-SAT<br>hours | ALL WEEK<br>hours | DAYS WORKED<br>Mon-Sat |
|-------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|-------------------|------------------------|
| 1     | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      |
| 2     | 14:30<br>23:25 | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | RD             | RD             | 30h 00m          | 38h 55m           | 4                      |
| 3     | 07:15<br>15:45 | RD             | 08:30<br>16:30 | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:45 | 40h 25m          | 48h 55m           | 5                      |
| 4     | RD             | RD             | 06:20<br>14:20 | 08:30<br>16:30 | RD             | 13:30<br>22:00 | 15:15<br>22:30 | 31h 45m          | 31h 45m           | 4                      |
| 5     | 14:30<br>23:25 | RD             | 14:00<br>22:30 | 14:00<br>22:30 | 14:00<br>22:30 | 16:25<br>23:55 | RD             | 33h 00m          | 41h 55m           | 4                      |
| 6     | RD             | 06:20<br>14:20 | 07:00<br>15:30 | 07:00<br>15:30 | 07:00<br>15:30 | 08:30<br>17:00 | RD             | 42h 00m          | 42h 00m           | 5                      |
| 7     | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      |
| 8     | 07:15<br>15:45 | 06:20<br>13:45 | RD             | RD             | 13:30<br>22:00 | 14:00<br>22:30 | 15:15<br>22:30 | 31h 40m          | 40h 10m           | 4                      |
| 9     | RD             | 12:00<br>20:30 | 12:00<br>20:30 | 12:00<br>20:30 | 12:00<br>20:30 | 12:00<br>20:30 | RD             | 42h 30m          | 42h 30m           | 5                      |
| 10    | RD             | 14:00<br>22:30 | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | 15:15<br>23:55 | 47h 10m          | 47h 10m           | 6                      |
| 11    | RD             | 07:00<br>15:30 | 07:00<br>15:30 | RD             | RD             | 06:20<br>13:35 | 06:20<br>14:45 | 32h 40m          | 32h 40m           | 4                      |
| 12    | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      |
| 13    | RD             | RD             | 14:00<br>22:30 | 13:30<br>22:00 | 14:00<br>22:30 | 14:00<br>22:30 | 15:15<br>23:55 | 42h 40m          | 42h 40m           | 5                      |
| 14    | RD             | 16:25<br>23:55 | 16:25<br>23:55 | RD             | RD             | 07:00<br>15:30 | 06:20<br>14:45 | 31h 55m          | 31h 55m           | 4                      |
| 15    | 07:15<br>15:45 | RD             | 06:20<br>13:35 | 06:20<br>13:35 | 06:20<br>13:35 | 07:00<br>15:30 | RD             | 30h 15m          | 38h 45m           | 4                      |
| 16    | RD             | 06:20<br>13:45 | 08:30<br>17:00 | 08:30<br>17:00 | RD             | RD             | RD             | 24h 25m          | 24h 25m           | 3                      |
| 17    | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      |
| 18    | 10:30<br>19:00 | 16:25<br>23:55 | RD             | RD             | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:45 | 31h 55m          | 40h 25m           | 4                      |
| 19    | 14:30<br>23:25 | RD             | 06:20<br>13:45 | 06:20<br>13:45 | 06:20<br>13:45 | 06:20<br>13:45 | 06:20<br>14:45 | 38h 05m          | 47h 00m           | 5                      |
| 20    | 08:30<br>16:30 | 06:20<br>13:45 | RD             | RD             | 13:30<br>22:00 | 16:25<br>23:55 | 15:15<br>23:55 | 32h 05m          | 40h 05m           | 4                      |
| 21    | 13:30<br>22:00 | 13:30<br>22:00 | RD             | 13:30<br>22:00 | 13:30<br>22:00 | 13:00<br>21:00 | RD             | 33h 30m          | 42h 00m           | 4                      |
| 22    | RD             | 14:00<br>22:30 | 13:30<br>22:00 | 13:30<br>22:00 | 13:30<br>22:00 | RD             | RD             | 34h 00m          | 34h 00m           | 4                      |
| 23    | RD             | 08:30<br>17:00 | RD             | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | 15:15<br>23:55 | 39h 40m          | 39h 40m           | 5                      |
| 24    | RD             | RD             | 06:20<br>13:45 | 06:20<br>13:45 | 07:00<br>15:30 | RD             | 09:30<br>16:30 | 30h 20m          | 30h 20m           | 4                      |
| Cover | 9<br>SP 4      | 13<br>SP 4     | 15<br>SP 4     | 15<br>SP 4     | 16<br>SP 4     | 16<br>SP 4     | 12<br>SP 4     | 35h 00m          | 38h 13m           | 4.35                   |

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines

# Same feel, more people

What this design keeps of today, and where the December headcounts differ

## Kept — the shape of a week

| MEASURE                                                  | TODAY (20)          | PROPOSED (24)              |
|----------------------------------------------------------|---------------------|----------------------------|
| Working weeks that are one turn (one clock time Mon–Fri) | 7 of 16             | <b>2 of 20</b>             |
| Weeks mixing early and late turns                        | 7                   | <b>8</b>                   |
| Rest-day breaks that are two days or more                | 13 of 17            | <b>15 of 27</b>            |
| Single rest days between duties (not a 48h break)        | 4                   | <b>12</b>                  |
| Days worked in a week (lines × days)                     | 1×3, 3×4, 11×5, 1×6 | <b>1×3, 5×4, 11×5, 3×6</b> |
| Cover (spare) weeks (whole weeks)                        | lines 1, 7, 12, 17  | <b>lines 1, 7, 12, 17</b>  |
| Distinct shift times (all already on today's roster)     | 18                  | <b>18</b>                  |
| Mon–Sat hours, average week (the contract)               | 35h 00m             | <b>35h 00m</b>             |
| Sunday duties (not contracted; half the working lines)   | 8 of 16 lines       | <b>9 of 20 lines</b>       |

## Changed — the December headcount

| PEOPLE ON DUTY                      | TODAY | PROPOSED  | WHY                                                             |
|-------------------------------------|-------|-----------|-----------------------------------------------------------------|
| Monday to Friday                    | 12    | <b>15</b> | the contract: 20 working lines × 35h has to be worked somewhere |
| Saturday                            | 10    | <b>12</b> | owner's figure for December, leaning late for events            |
| Sunday                              | 8     | <b>9</b>  | owner's figure for December                                     |
| Opening at 06:20 (07:15 Sunday)     | 4     | <b>4</b>  | unchanged                                                       |
| Through to the 23:55 close, weekday | 2     | <b>3</b>  | three to close                                                  |
| Through to the close, Saturday      | 2     | <b>4</b>  | four on a Saturday                                              |
| Still on duty at 22:00              | 4     | <b>5</b>  | five are enough from 22:00                                      |

## The duty table, in today's times

| TURN                                  | WK | SAT | SUN  | WK | SAT  | SUN  |
|---------------------------------------|----|-----|------|----|------|------|
| 06:20–13:35 7h 15m · Early            | 1  |     | 1    |    |      |      |
| 06:20–13:45 7h 25m · Early            | 1  |     | 3 +2 |    |      |      |
| <del>06:20–14:00 7h 40m · Early</del> |    | 4   |      |    | −4   |      |
| 06:20–14:20 8h 00m · Early            | 2  |     | 2    |    |      |      |
| 06:20–14:45 8h 25m · Early            |    |     |      |    | 5 +5 |      |
| <del>06:20–14:50 8h 30m · Early</del> |    | 3   |      |    | −3   |      |
| 07:00–15:30 8h 30m · Early            |    |     | 2 +2 |    |      |      |
| 07:15–15:45 8h 30m · Early            |    |     | 3    |    |      | 3    |
| <del>08:00–16:20 9h 30m · Early</del> | 2  |     | −2   |    |      |      |
| 08:30–16:30 8h 00m · Early            |    |     | 1 +1 |    |      | 1    |
| 08:30–17:00 8h 30m · Early            |    |     | 1 +1 |    |      |      |
| 09:30–16:30 7h 00m · Early            |    |     |      |    | 1 +1 |      |
| 10:30–19:00 8h 30m · Early            |    |     |      |    |      | 1 +1 |
| <del>11:00–19:30 9h 30m · Late</del>  | 4  |     | −4   |    |      |      |
| <del>12:00–20:00 9h 00m · Late</del>  |    | 4   |      |    | −4   |      |
| 12:00–20:30 8h 30m · Late             |    |     | 1 +1 |    |      |      |
| 13:00–21:00 8h 00m · Late             |    |     | 1 +1 |    |      | −1   |
| <del>13:30–21:00 7h 30m · Late</del>  |    | 4   |      |    | −4   |      |
| 13:30–22:00 8h 30m · Late             | 1  |     | 4 +3 |    |      | 1 +1 |
| 14:00–22:30 8h 30m · Late             | 2  |     | 2    |    |      |      |
| <del>14:30–22:00 7h 30m · Late</del>  |    | 2   |      |    | −2   |      |
| 14:30–23:25 8h 55m · Late             |    |     | 3    |    |      | 3    |
| <del>14:45–23:55 9h 10m · Late</del>  |    | 2   |      |    | −2   |      |
| 15:15–22:30 7h 15m · Late             |    |     |      |    | 2 +2 |      |
| 15:15–23:55 8h 40m · Late             | 2  |     | −2   |    | 4 +4 |      |
| 16:25–23:55 7h 30m · Late             |    |     | 4 +4 |    |      |      |

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the duties land where they do: Monday to Saturday totals exactly 42,000 minutes, which is 20 × 35h to the minute, so the contract is paid. But unlike a table built day-uniform, **the weekdays here are not equal** — Mon 6,225 · Tue 7,205 · Wed 7,205 · Thu 7,780 · Fri 7,690, and Saturday 5,895. Monday is the lightest day by some margin and Thursday the heaviest; that is a property of the design rather than a fault, but it is the first thing to put to the roster office, because it decides how much cover each day really has.

# People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

## Monday to Friday window 06:20–23:55

|                  | 05 | 06 | 07  | 08  | 09  | 10 | 11 | 12 | 13 | 14 | 15 | 16  | 17  | 18  | 19  | 20 | 21 | 22 | 23 |
|------------------|----|----|-----|-----|-----|----|----|----|----|----|----|-----|-----|-----|-----|----|----|----|----|
| TODAY            |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 8  | 8  | 8  | 8   | 6   | 6   | 6   | 5  | 5  | 4  | 2  |
| PROPOSED         |    | 4  | 6   | 8   | 8   | 8  | 8  | 9  | 10 | 9  | 8  | 9   | 7   | 7   | 7   | 7  | 6  | 5  | 3  |
| DEC 2026 TRAFFIC | 5  | 69 | 107 | 127 | 122 | 95 | 85 | 75 | 84 | 79 | 87 | 105 | 140 | 138 | 107 | 90 | 73 | 75 | 66 |

## Saturday window 06:20–23:55

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| TODAY            |    | 4  | 4  | 4  | 4  | 4  | 4  | 5  | 6  | 9  | 6  | 6  | 6  | 6  | 6  | 5  | 4  | 2  | 2  |
| PROPOSED         |    | 5  | 5  | 5  | 6  | 6  | 6  | 6  | 6  | 6  | 7  | 7  | 6  | 6  | 6  | 6  | 6  | 6  | 4  |
| DEC 2026 TRAFFIC | 6  | 24 | 58 | 75 | 81 | 92 | 88 | 77 | 64 | 68 | 72 | 84 | 82 | 67 | 64 | 79 | 78 | 62 | 41 |

## Sunday window 07:15–23:25

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| TODAY            |    |    | 3  | 4  | 4  | 4  | 4  | 4  | 5  | 8  | 8  | 5  | 4  | 4  | 4  | 4  | 3  | 3  | 3  |
| PROPOSED         |    |    | 3  | 4  | 4  | 5  | 5  | 5  | 6  | 9  | 9  | 6  | 5  | 5  | 4  | 4  | 4  | 3  | 3  |
| DEC 2026 TRAFFIC |    |    | 15 | 45 | 65 | 71 | 74 | 71 | 61 | 55 | 64 | 69 | 78 | 73 | 75 | 52 | 71 | 65 | 50 |

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. Sunday: 5 December movements fall after the 23:25 finish — the standing question on whether Sunday's window moves; the window is stored per design, so this proposal can be rebuilt to either answer.

**Reading it.** The proposal raises the whole day rather than reshaping it, which is what the December brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The one hour the search was allowed to favour is Saturday's 17:00–22:00, for events. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

# The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

**Hard limits** — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **9** days (today: 9)  
Longest possible run is 9 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **9** days — so this row does not depend on how a cover week is placed.  
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)  
The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks  
700h of duty a week across 20 working lines. Sundays (77.25h) sit on top as RDW, as they do today. Individual weeks range 24h 25m–47h 10m; only the average is the contract.

**December design figures** — the staffing shape agreed for the new timetable

| RULE                                           | PROPOSAL                                                                                              |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Four on at the open, every day                 | × 4 weekday · 5 Saturday · 3 Sunday                                                                   |
| Three through to the close; four on a Saturday | ✓ 3 weekday · 4 Saturday · 3 Sunday                                                                   |
| Five still on duty at 22:00                    | × 5 weekday · 6 Saturday      a 22:00 finish is not "on at 22:00"                                     |
| Fourteen on a Saturday, ten on a Sunday        | × 12 · 9                                                                                              |
| Cover weeks, evenly spread                     | × 4 weeks at lines 1, 7, 12, 17 — gaps 8, 6, 5, 5      the December shape asks for five               |
| About 4.2 days a week worked, Mon-Sat          | ✓ 4.35 over the 20 working lines                                                                      |
| No :05 or :10 times except the open and close  | ✓ none                                                                                                |
| Late turns slightly shorter than most earlies  | × not met — lates 7h15–8h40, earlies 7h00–8h30      the longest late is longer than the longest early |

## Two things to settle before it is frozen

**Saturday was running flat against a curve that is not.** Cover sat at six people for most of the day — the same six at 10:00, when demand peaks at 92 cars, as at 13:00, when it falls to 64. **Nine turns become four, and six start times become three:** 06:20–14:45 five times, 09:30–16:30 once, and 15:15 starting BOTH late turns — 15:15–22:30 twice, off when the ticket office closes, and 15:15–23:55 four times, through to the station close. Finishes fall from seven to four. The thinnest fully-covered hour rises from **3.0 people to 4.7**, and the longest Saturday duty falls from **9h10 to 8h40**, so Saturday now sits under the cap the rest of this folder works to. **The late carries a little more, on purpose.** Against the measured curve alone the fit is 11.4 where today's Saturday is 20.7; against the Wembley-weighted target it is 11.8 where today's is 32.8. The weight is deliberately gentle and it buys half a person at 16:00 and half at 22:00, paid for by half at 08:00 and half at 09:00. A heavier weight was tried and rejected: at 1.25 it buys a whole person at 16:00 and gives a whole one back at 09:00, where the curve reads 81 cars. 1.10, 1.15 and 1.20 all return this same table, so the answer does not balance on the knob. **One December rule is now met that was not.** Four people through to the close, against three before. **One is not, and cannot be.** The rule asks for four at the open and Saturday still has five. That is not a search failure: no table of 12 duties paying 5,895 minutes, drawn from this pool, puts four at the open AND four at the close, at any evening floor. Four at the open needs either a different Saturday headcount or minutes moved in from a weekday, and both are outside what this revision was allowed to touch. **What it cost.** The worst rolling seven-day total rises from **53.1 hours to 53.5**, still well under 55. Every other figure is unchanged — three factors present on both readings, FF11 at 13, the nine-day run, five weekends off, no rest under 12 hours and no hard-limit breach. Saturday's headcount is still **12 against the December shape's fourteen**, and that is the real limit on how much shape Saturday can carry: 5,895 minutes across a 17h35 day is a mean of 5.6 people, so there is little headroom to build a peak whatever the table.

# ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021).  $\Delta$  present · ✓ clear · ● standing, a property of the operation · – not applicable. Nathan assesses against this list.

| CODE | FACTOR                                                                                | TODAY'S LINK                                | PROPOSED                                     |
|------|---------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------|
| FF1  | Night shift covering 00:00–05:00 Time of day                                          | – –                                         | – –                                          |
| FF2  | Early shift starting 05:00–07:00 Time of day                                          | ● 24                                        | ● 25                                         |
| FF3  | Very early shift starting before 05:00 Time of day                                    | ✓ 0                                         | ✓ 0                                          |
| FF4  | Very early shift (before 05:00) over 8h Duty length                                   | – –                                         | – –                                          |
| FF5  | Day shift over 12h Duty length                                                        | ✓ 0                                         | ✓ 0                                          |
| FF7  | Early shift over 10h Duty length                                                      | ✓ 0                                         | ✓ 0                                          |
| FF8b | Fewer than 2 days rest after a block of early starts Recovery time                    | $\Delta$ 3                                  | $\Delta$ 5                                   |
| FF11 | More than 13 consecutive shifts without a 48h break Recovery time                     | ✓ 12<br>✓ 9 as rostered                     | ✓ 13<br>✓ 11 as rostered                     |
| FF10 | More than 4 consecutive 12h day shifts Cumulative                                     | ✓ 0                                         | ✓ 0                                          |
| FF15 | More than 4 consecutive early shifts in a rotating pattern Cumulative                 | $\Delta$ 6                                  | $\Delta$ 6                                   |
| MRSF | More than 12 consecutive day shifts Cumulative                                        | ✓ 9                                         | ✓ 9                                          |
| MRSF | More than 55 hours worked in any 7-day period Cumulative                              | $\Delta$ 60.8                               | ✓ 53.5                                       |
| MRSF | More than 7 consecutive 8h shifts (definition to confirm) Cumulative                  | $\Delta$ 9                                  | ✓ 5                                          |
| MRSF | More than 6 consecutive night or early shifts in a permanent pattern Cumulative       | – –                                         | – –                                          |
| FF17 | Backward rotating pattern (definition to confirm) Circadian                           | ✓ 10 backward / 12 forward                  | ✓ 12 backward / 19 forward                   |
| FF18 | Rotating pattern of about a week (definition to confirm) Circadian                    | ● 20-line rotation · typically 4h 0m a week | ● 24-line rotation · typically 3h 54m a week |
| FF19 | Successive start times varying by more than 2 hours (definition to confirm) Circadian | $\Delta$ 6                                  | $\Delta$ 10                                  |
| FF6  | Night shift over 10h Night shifts                                                     | – –                                         | – –                                          |
| FF8  | Less than 2 days rest after consecutive nights Night shifts                           | – –                                         | – –                                          |
| FF9  | Less than 14h rest in any 24h for night shifts Night shifts                           | – –                                         | – –                                          |
| FF12 | Only one day rest after night shifts Night shifts                                     | – –                                         | – –                                          |
| FF14 | More than 4 consecutive nights in a rotating pattern Night shifts                     | – –                                         | – –                                          |
| FF16 | More than 3 consecutive night shifts over 8h Night shifts                             | – –                                         | – –                                          |
| FF20 | First night shift Night shifts                                                        | – –                                         | – –                                          |
| FF13 | Less than 12h rest in any 24h Intervals between duties                                | ✓ 0                                         | ✓ 0                                          |

Present: 3 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

# Where it came from, and how to load it

The Saturday table was searched; everything else is carried over unchanged from Cover at Seventeen

## Where this design came from

**1 • Saturday only, and that is checkable.** Against *Cover at Seventeen*, **zero** cells outside the Saturday column differ. Saturday still has **12 people** and still pays **5,895 minutes**, so Monday to Saturday is 42,000 minutes against 20 working lines × 35h — exactly, as before. The contract could not move, because the budget the search was given was the one Saturday already had.

**2 • The table was searched, not drawn.** Every multiset of 12 duties paying exactly 5,895 minutes was tried, from a pool of quarter-hour starts and finishes (plus the 06:20 open and the 23:55 close), each duty between 7h00 and 8h40 — about 130 million tables, from a SEEDED search, so it reproduces. Two rules from the owner are built into the pool rather than applied afterwards: in the 21:00–23:00 band the only legal finish is **22:30**, because that is when the ticket office closes, and the demand TARGET for 15:00–23:00 is lifted 15% for **Wembley** — Chiltern serves Wembley Stadium out of Marylebone, and an event fills trains the timetable curve already counts. They were scored by the same demand measure the other proposals use, against the measured December Saturday curve, with fewer distinct turns as the tie-break and a floor under the thinnest hour so nothing could be won by hollowing out the evening.

### 3 • Then the weeks were moved, on Saturday alone.

Which line holds which Saturday duty was searched exhaustively under the 12-hour rest rule — Friday into Saturday on the same line, and Saturday into the next line's Sunday. 90 assignments clear it; the one printed is the best of them on the fatigue figures.

### 4 • The judge is the workspace's own —

runDesignChecks, weeklyHours, assessHardLimits, assessFatigue, scoreOrder and calcHourlyCoverage, with nothing typed by hand.

## Paste it into the workspace

Links → Import, then paste the block on page 8. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied as proposal-import.txt (tab-separated, pastes directly) and proposal.json (the app's own format).

## Alternatives measured

| DESIGN                                                                       | RUN | FF PRESENT | WKENDS | ONE-TURN | STEP   | WK FIT | SCORE |
|------------------------------------------------------------------------------|-----|------------|--------|----------|--------|--------|-------|
| S4-24-EXT<br>· 481ba9ed<br>—<br><b>Saturday<br/>Four (this<br/>proposal)</b> | 9   | 3          | 5      | 2/20     | 3h 54m | 46.9   | —     |
| Same<br>Turns ·<br>d15e1b74<br>— the<br>searched<br>proposal                 | 6   | 1          | 6      | 18/20    | 2h 33m | 58.5   | —     |
| By the<br>Book ·<br>0f14abce<br>— the<br>searched<br>proposal                | 6   | 0          | 6      | 12/20    | 2h 52m | 41.1   | —     |
| C17-24-<br>EXT —<br>before<br>Saturday<br>was rebuilt<br>· edc1b731          | 9   | 3          | 5      | 2/20     | 3h 57m | 46.9   | —     |
| WS-24-<br>EXT —<br>where the<br>line started<br>· 0bebb675                   | 9   | 5          | 6      | 6/20     | 4h 17m | 46.9   | —     |
| WL4-24-<br>F7 — the<br>same work,<br>searched ·<br>f0d403d6                  | 7   | 1          | 6      | 10/20    | 2h 34m | 46.9   | —     |
| Workspace<br>default ·<br>29ec9810<br>(Dec 2026<br>table,<br>generated)      | 7   | 5          | 5      | 0/20     | 4h 16m | 41.1   | —     |
| Today's<br>20-line link<br>(for scale)                                       | 9   | 5          | 4      | 7/16     | 4h 0m  | 57.7   | —     |

**How the winner was picked:** rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the December traffic curve (lower is better; today's link scores what it scores). Two candidates tied on every rule; the fit decided it, and the runner-up is named so it can be asked for. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

LOAD IT

## The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

|    |             |             |             |             |             |             |             |
|----|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 1  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 2  | 14:30-23:25 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | RD          | RD          |
| 3  | 07:15-15:45 | RD          | 08:30-16:30 | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | 06:20-14:45 |
| 4  | RD          | RD          | 06:20-14:20 | 08:30-16:30 | RD          | 13:30-22:00 | 15:15-22:30 |
| 5  | 14:30-23:25 | RD          | 14:00-22:30 | 14:00-22:30 | 14:00-22:30 | 16:25-23:55 | RD          |
| 6  | RD          | 06:20-14:20 | 07:00-15:30 | 07:00-15:30 | 07:00-15:30 | 08:30-17:00 | RD          |
| 7  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 8  | 07:15-15:45 | 06:20-13:45 | RD          | RD          | 13:30-22:00 | 14:00-22:30 | 15:15-22:30 |
| 9  | RD          | 12:00-20:30 | 12:00-20:30 | 12:00-20:30 | 12:00-20:30 | 12:00-20:30 | RD          |
| 10 | RD          | 14:00-22:30 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 15:15-23:55 |
| 11 | RD          | 07:00-15:30 | 07:00-15:30 | RD          | RD          | 06:20-13:35 | 06:20-14:45 |
| 12 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 13 | RD          | RD          | 14:00-22:30 | 13:30-22:00 | 14:00-22:30 | 14:00-22:30 | 15:15-23:55 |
| 14 | RD          | 16:25-23:55 | 16:25-23:55 | RD          | RD          | 07:00-15:30 | 06:20-14:45 |
| 15 | 07:15-15:45 | RD          | 06:20-13:35 | 06:20-13:35 | 06:20-13:35 | 07:00-15:30 | RD          |
| 16 | RD          | 06:20-13:45 | 08:30-17:00 | 08:30-17:00 | RD          | RD          | RD          |
| 17 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 18 | 10:30-19:00 | 16:25-23:55 | RD          | RD          | 06:20-14:20 | 06:20-14:20 | 06:20-14:45 |
| 19 | 14:30-23:25 | RD          | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | 06:20-14:45 |
| 20 | 08:30-16:30 | 06:20-13:45 | RD          | RD          | 13:30-22:00 | 16:25-23:55 | 15:15-23:55 |
| 21 | 13:30-22:00 | 13:30-22:00 | RD          | 13:30-22:00 | 13:30-22:00 | 13:00-21:00 | RD          |
| 22 | RD          | 14:00-22:30 | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | RD          | RD          |
| 23 | RD          | 08:30-17:00 | RD          | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 15:15-23:55 |
| 24 | RD          | RD          | 06:20-13:45 | 06:20-13:45 | 07:00-15:30 | RD          | 09:30-16:30 |

Prepared 22 September 2026. Every figure in this document was computed by the Marylebone Roster app's Links modules from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.